

TOBY LAU

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EDUCATION

University of British Columbia

B.Sc. Computer Science – GPA: 3.95

May 2028 (Expected)

Vancouver, BC

EXPERIENCE

Squarepoint Capital

Montreal, QC

Software Engineer Intern – Data Services (Portfolio Construction & Trading)

Sep 2026 – Dec 2026

- Building a Barra risk analytics platform for PMs to decompose risk and PnL across portfolios, trade baskets, and strategies.
- Introduced qunit snapshot tests for a kdb+/q web service using recorded live responses as mocks and expected output CSVs.
- Migrated **17** microservices into **2** devcontainer-based monorepos, saving **40+** engineering hours in setup/onboarding time.

Software Engineer Intern – Data Services (Commodities)

May 2026 – Aug 2026

- Created a monitoring platform for trading desks to schedule, aggregate, and set up alerts on health checks for **1100+** alphas.
- Cut p95 latency **22%** on **35+** prod GraphQL operations by folding a gRPC layer into an Apollo subgraph to drop a network hop.
- Reduced data read per time-series query **~72%** across **100+** strategies using adaptive time ranges instead of fixed lookbacks.
- Built a distributed poller/worker data pipeline with Python ARQ to ingest, process, and store **~20k** alpha metrics per second.
- Designed a YAML-database sync mechanism enabling **50+** desks to version-control dashboard configurations through GitLab.

Planview

Vancouver, BC

Software Engineer Intern – Planview Copilot

Jan 2025 – Aug 2025

- Reduced chat latency **50%+** by batching independent LLM tool calls into parallel AWS Lambda invocations in a Java service.
- Unblocked a major product release for **2k+** users by implementing a dual-token handler during an auth service migration.
- Implemented a log obfuscator/forwarder that routed **21k+** sanitized daily logs to Datadog, retaining raw logs in CloudWatch.
- Delivered a self-service RAG platform for **100+** tenants with S3 and Bedrock, enabling document uploads and analysis/Q&A.
- Prototyped a hybrid tool-calling/MCP LLM chat service and registry to support **8** existing tools while integrating MCP servers.

UBC Department of Computer Science

Vancouver, BC

Undergraduate Teaching Assistant – CPSC 210 (Introduction to Object Oriented Programming)

Sep 2024 – Dec 2024

- Taught OOP/OOD, design patterns, and TDD with JUnit to **300+** students; mentored **25+** students in weekly labs/office hours.

APL Zuellig Pharma

Remote

Software Engineer Intern – Distribution Resource Planning

Oct 2021 – Jan 2022

- Saved **10+** hrs/week by replacing an Excel stock forecasting workflow with a Python GUI for **46k+** SKUs across **25** warehouses.

EXTRACURRICULARS

Databricks

Remote

Student Fellow

Jun 2026 – Present

- **1 of 36** selected from **5k+** globally to represent Databricks campus/user groups, with mentorship from Databricks engineers.

PROJECTS

Extreme Vocals Classifier | Python, PyTorch, NumPy, SciPy, Librosa, Google Colab

- Synthesized a **500+** extreme vocal sample dataset by isolating vocals with Demucs, extracting and processing relevant vocal regions with NumPy/SciPy (RMS normalization, high-pass filter), and augmenting vocals with librosa from **57+** songs.

Agent Eval RL Environment + Trajectory Viewer | React, TypeScript, Python, OpenRouter, Harbor Framework

- Built an RL environment with mock Jira/Slack tools and designed multi-step tasks for agentic evals using Harbor Framework.

Membership Portal | React, TypeScript, PostgreSQL, Stripe, Supabase, Google Cloud Platform

- Supported **250+ MAU** and **\$5k+** revenue growth by building payments, registration flows, and admin tools from **0**→**1**.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, HTML, CSS, SQL, C, C++, kdb+/q

Frameworks: React, Spring Boot, Next.js, Express, Flask, FastAPI, JUnit, Jest

Technologies: AWS, GCP, Supabase, PostgreSQL, Docker, Kubernetes, ArgoCD, Liquibase/Alembic, GraphQL, gRPC

Tools: Git, Postman/Insomnia, Datadog, Elasticsearch, Jupyter, DBeaver, Vim, Figma, Jira, IntelliJ, Claude Code